

Northwind Traders Sales and Business Insights Analysis

P.Moksha Manisha

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1 Business Objective

The objective of this project is to analyze the operational data of Northwind Traders to extract meaningful business insights. Using data visualization and exploratory data analysis techniques, the project investigates customer behavior, product sales patterns, employee distribution, and supplier relationships. The goal is to support business decision-making through data-driven insights and interactive visual dashboards created using Power BI.

2 Dataset Overview

The Northwind dataset represents a fictional wholesale trading company that imports and exports specialty foods from different parts of the world. The dataset contains several relational tables that describe different aspects of the company's operations.

Key Tables Used in the Analysis:

- Customers – Stores information about customer companies and locations.
- Orders – Contains records of all orders placed by customers.
- Order Details – Provides detailed information about products in each order including quantity and price.
- Products – Contains product information including price, stock, and supplier.
- Categories – Defines product categories.
- Employees – Stores employee details such as title, hire date, and region.
- Suppliers – Contains supplier company and geographic information.
- Shippers – Contains shipping company information.

These tables are connected through primary and foreign keys such as *CustomerID*, *OrderID*, *ProductID*, and *SupplierID*, enabling integrated analysis across multiple business dimensions.

3 MECE Analytical Framework

To ensure structured and non-overlapping analysis, the project follows a MECE (Mutually Exclusive, Collectively Exhaustive) framework. The analysis is divided into four major categories:

3.1 Customer Analysis

- Average number of orders per customer
- Customer order distribution by country and city
- Customer segmentation based on order frequency and spending
- Identification of high-value repeat customers

3.2 Product and Sales Analysis

- Identification of products and categories contributing the highest revenue
- Monthly and seasonal demand patterns
- Correlation between product price, stock levels, and revenue
- Detection of anomalies in product sales performance

3.3 Employee Analysis

- Geographic distribution of employees
- Job title distribution and workforce structure
- Hiring trends over time based on employee hire dates

3.4 Supplier Analysis

- Geographic distribution of suppliers
- Supplier participation across different product categories
- Regional pricing trends across suppliers

4 Exploratory Data Analysis (EDA) Insights

The exploratory data analysis phase focused on identifying patterns and relationships within the dataset.

- The analysis identified customers who place orders frequently, highlighting the presence of repeat and high-value customers.
- Customer order patterns vary across different countries and cities, indicating strong market regions for the company.
- Product categories contribute unevenly to overall revenue, with certain products generating significantly higher sales.
- Employee distribution shows variations across job titles and geographic regions, reflecting organizational structure.
- Product demand changes across months, indicating seasonal sales trends that can affect inventory planning.
- Some products demonstrate unusually high or low sales performance, helping identify sales anomalies.
- Supplier distribution varies across regions, showing that certain regions provide more products to the company.

5 Power BI Dashboard Analysis

An interactive Power BI dashboard was developed to visualize the key findings of the analysis. The dashboard includes multiple report pages that represent different analytical perspectives.

The dashboard visualizes:

- Customer distribution across countries and cities
- Order trends over time
- Customer segmentation by contact titles
- Order value distribution
- Employee distribution by job roles
- Product sales performance

- Supplier distribution and pricing analysis

These visualizations allow stakeholders to interact with the data and explore business patterns dynamically.

6 Key Business Insights

From the combined EDA and dashboard analysis, several important insights were identified:

- A small group of customers contributes a significant share of the total orders.
- Certain geographic regions generate more customer orders compared to others.
- A few product categories dominate revenue generation.
- Seasonal patterns influence product demand across different months.
- Supplier pricing varies across different regions and categories.

7 Business Recommendations

Based on the insights derived from the analysis, the following recommendations can support business improvements:

- Focus marketing strategies on high-value repeat customers.
- Maintain optimal inventory levels for high-demand products.
- Investigate underperforming products to improve sales performance.
- Strengthen supplier partnerships in regions that provide consistent supply.
- Use seasonal demand patterns to improve supply chain planning.

8 Conclusion

This project demonstrates how data analytics and visualization techniques can transform raw business data into actionable insights. By analyzing customer behavior, product performance, employee structure, and supplier relationships, the project highlights opportunities for improving operational efficiency and strategic decision-making at Northwind Traders. The Power BI dashboard provides a user-friendly platform for stakeholders to explore these insights interactively.